Implementation Roadmap

Phase-wise Delivery Plan & Milestones

Project: Solar MES Implementation — 300 MW annually | Total Duration: 20.5 months | 02 May 2026

1. Programme Summary

Phase	Name	Duration	Weeks	Key Focus
Phase 0	Pre-Project Assessment	4 weeks	1–4	Machine connectivity audit; SAP landscape; data quality; business case sign-off
Phase 1	Discovery & Blueprint	8 weeks	5–12	AS-IS mapping; FSD; integration design; project charter; risk register
Phase 2	Core MES Configuration	16 weeks	13–28	MES setup; QMS; genealogy; SAP integration; SFDC; unit testing
Phase 3	Equipment Integration	16 weeks	17–32	OPC UA; Kafka; machine adapters (flash, EL, laminator, stringer); edge devices
Phase 4	PODISE / APS Deployment	10 weeks	21–30	Scheduling engine; Gantt portal; lot splitting; SAP order dressing; parallel run
Phase 5	Testing & Go-Live	8 weeks	29–36	UAT; load/DR/security test; pilot line go-live; 4-week hypercare
Phase 6	Expansion & Optimisation	16 weeks	37–52	Roll to all lines; AI model training; ALMM automation; advanced analytics

2. High-Level Gantt (52-Week View)

Each block represents 4 weeks. ■ = active, ■ = not started, ■ = go-live milestone.

Phase / Activity	Q1 (W1–12)	Q2 (W13–24)	Q3 (W25–36)	Q4 (W37–52)
Phase 0: Pre-Project	■■■■	■■■■	■■■■	■■■■
Phase 1: Blueprint	■■■■	■■■■	■■■■	■■■■
Phase 2: MES Config	■■■■	■■■■	■■■■	■■■■
Phase 3: Equip. Integr.	■■■■	■■■■	■■■■	■■■■
Phase 4: PODISE/APS	■■■■	■■■■	■■■■	■■■■
Phase 5: Test & Go-Live ■	■■■■	■■■■	■■■■	■■■■
Phase 6: Expansion	■■■■	■■■■	■■■■	■■■■

3. Phase-by-Phase Detail

Phase 0: Pre-Project Assessment — Weeks 1–4

- Factory walk-through: all production lines, machines, network points
- Machine connectivity audit: OPC UA readiness, PLC type, data availability per machine
- SAP landscape assessment: version (ECC/S4), active modules, integration options
- Data quality assessment: BOM completeness, routing accuracy, work center master
- IT infrastructure review: server room, network topology, security policy
- Business case finalisation: ROI model reviewed with CFO/COO

Deliverables: Connectivity audit report | SAP assessment report | Infrastructure specification | Approved business case | Go/no-go decision
Go/No-Go Criteria: Business case approved; budget allocated; project sponsor confirmed

Phase 1: Discovery & Blueprint — Weeks 5–12

- AS-IS process mapping: all 11 solar manufacturing steps documented
- Gap analysis: current state vs. target state per process area
- Functional Specification Document (FSD): all MES modules, 200+ pages
- Integration design specification: SAP (IDoc/REST), OPC UA (tag list per machine type), WMS
- Data migration plan: master data from SAP/Excel, cleansing rules
- Project charter, RACI matrix, risk register, change management plan
- Training needs assessment: 60 operators + supervisors + IT team

Deliverables: AS-IS process maps | FSD (signed by all stakeholders) | Integration design spec | Project charter | Risk register
Go/No-Go Criteria: FSD signed by Plant Manager, IT Manager, Quality Manager, and SAP team

Phase 2: Core MES Configuration — Weeks 13–28

- Work center, BOM, routing master data loading (from SAP export + cleansing)
- Production order management configuration: lifecycle, status, priority rules
- Shop floor UI design: operation-specific screens, barcode/tablet entry
- QMS configuration: inspection plans, defect code library (IEC 61215 mapped), SPC charts
- Material genealogy: lot traceability rules, serial number format, ALMM compliance fields
- SAP integration: IDoc production order receipt, confirmation posting, GR/GI
- OEE dashboard: Availability/Performance/Quality metrics, downtime reason codes
- Unit testing: all 200+ test cases from FSD; defect resolution

Deliverables: Configured MES environment | SAP integration tested | OEE dashboards live | QMS configured | Unit test report
Go/No-Go Criteria: All unit tests passed; SAP round-trip test completed; UAT plan approved

Phase 3: Equipment Integration — Weeks 17–32

- OPC UA tag mapping: all 20 machine types mapped to MES data model
- Edge IoT gateway deployment: 1 per line, network config, OT VLAN setup
- Kafka topic design: PDI/PDO, quality results, alarms, maintenance events
- Flash tester integration: IV curve (Pmax, Voc, Isc, FF) per module serial
- EL tester integration: AI image classification deployment, defect type mapping

• Laminator temperature profile capture: set points vs. actuals per cycle

• Stringer AOI integration: string count, ribbon consumption, solder quality

• Interface testing: machine data vs. manual record validation (±2% tolerance)

Deliverables: All machines connected and validated | Kafka event flows verified | AI EL classifier deployed | Interface test report
Go/No-Go Criteria: Machine data validated against manual records; ±2% tolerance achieved for all machines

Phase 4: PODISE / APS Deployment — Weeks 21–30

• Work center capacity model: shift calendar, break times, planned maintenance windows

• Lot-size constraint configuration: laminator (max 12), flash tester (1), packaging (30)

• Scheduling algorithm calibration: standard cycle times from Phase 3 actual data

• SAP order dressing integration: VA01 sales order → PODISE → CO01 production order

• Gantt dashboard deployment: Angular real-time; drag-and-drop override capability

• Parallel run: PODISE schedule vs. manual planner (4 weeks); accuracy measurement

• AI recommendations: plain-English rescheduling advice on rush orders / breakdowns

Deliverables: PODISE live on all work centers | Gantt portal deployed | Parallel run report (>95% accuracy target) | Planner training complete
Go/No-Go Criteria: PODISE schedule accuracy >95% vs. actual production; planner sign-off

Phase 5: Testing & Go-Live — Weeks 29–36

• User Acceptance Testing (UAT): super-users test all FSD scenarios

• Load testing: simulate peak day (120% of normal volume); performance validated

• Disaster recovery test: failover to standby server; RTO measurement

• Security penetration test: OT/IT segregation validated; auth tested

• Parallel run: old system + MES side by side for 2–4 weeks on pilot line

• Go-live: Line 1 pilot; 24/7 hypercare support for 4 weeks

• OEE baseline captured: before/after comparison for business case validation

Deliverables: UAT sign-off | Load test report | DR test report | Pilot line go-live | OEE baseline report
Go/No-Go Criteria: Plant Manager sign-off on UAT; OEE baseline captured; no P1 open defects

Phase 6: Expansion & Optimisation — Weeks 37–52

• Roll-out to remaining production lines (Lines 2, 3, 4) — 3-week each

• AI model fine-tuning: EL classifier trained on plant-specific defect images

• PODISE algorithm calibration: actual cycle times replace estimates

• ALMM compliance module: serial registration, COC auto-generation live

• Predictive maintenance: sensor baselines established; anomaly detection active

• Advanced analytics: yield correlation, supplier quality, energy efficiency

• Knowledge transfer: internal team capable of L2 support without vendor

Deliverables: All 4 lines live | ALMM COC automation | AI models optimised | Knowledge transfer complete | Final project report
Go/No-Go Criteria: All KPIs at or above target; internal team certified; project formally closed

4. RACI Matrix

Activity	Vendor PM	Vendor Tech	Plant Mgr	IT Mgr	Production	Quality	SAP Team
Business case	C	I	R/A	C	C	C	I
Requirements (FSD)	R	C	A	C	C	C	C
MES Configuration	R/A	R	I	C	C	C	I
SAP Integration	C	R	I	C	I	I	R/A
OPC UA Setup	R/A	R	I	C	I	I	I
UAT Execution	C	C	A	C	R	R	C
Training Delivery	R/A	C	I	I	C	C	I
Go-Live Decision	C	C	R/A	C	C	C	C
ALMM Compliance	C	R	I	I	I	R/A	I

R=Responsible, A=Accountable, C=Consulted, I=Informed

5. Key Project Milestones

Milestone	Target Week	Success Criteria	Owner
M0: Project Kick-off	Week 1	All stakeholders aligned; project plan signed	Vendor PM
M1: FSD Sign-off	Week 12	All 200+ requirements approved by all stakeholders	Plant Mgr
M2: Integration Design Approved	Week 12	SAP, OPC UA, Kafka specs approved by IT & SAP teams	IT Mgr
M3: MES Unit Test Complete	Week 28	All test cases passed; zero P1 defects open	Vendor PM
M4: Machine Integration Validated	Week 32	All machines data-validated; ±2% tolerance met	Vendor Tech
M5: PODISE Parallel Run Complete	Week 30	>95% scheduling accuracy vs. actual	Production
M6: UAT Sign-off	Week 34	All super-users sign off; zero P1 defects	Plant Mgr
M7: Pilot Line Go-Live ■	Week 36	Line 1 live; OEE visible; no critical issues	Plant Mgr
M8: All Lines Live	Week 50	All 4 lines operational; ALMM COC automated	Plant Mgr
M9: Project Closure	Week 52	KPIs validated; knowledge transfer complete	Vendor PM